

MindKeeper: A Dual-Brain AI Agent for Dementia Homecare

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Abstract

Dementia care requires an AI to perform two conflicting tasks: display deep empathy for the patient and provide strict, objective medical advice to the caregiver. A single LLM usually fails to do both simultaneously without hallucination. In this midterm project, we present MindKeeper, a Dementia Homecare Companion and Clinical Triage Agent. We built a Dual-Brain Architecture that separates workloads between a fine-tuned LoRA model for empathetic conversations and a base Qwen2.5-3B model for clinical logic. Our system comforts patients using Validation Therapy while generating safe action checklists for caregivers. This report details our system design, fine-tuning setup, and quantitative stress testing results.

1 Introduction

Managing Alzheimer’s disease and other types of dementia is difficult. Patients frequently suffer from delusions, hallucinations, and severe memory loss. When an AI agent talks to them, it must be highly empathetic. We rely on Validation Therapy in our system design. This means the AI should not correct the patient’s false reality, which helps reduce their anxiety.

However, if an AI is overly empathetic, it might agree to dangerous requests and give unsafe medical advice. This creates a critical alignment conflict. To address this problem, we engineered MindKeeper.

Role of AI Agent: Dementia Homecare Companion and Clinical Triage Agent.

Project Links:

- **GitHub Repository:** <https://github.com/Yiwen543/6895-midterm-project.git>
- **Demo Video:** https://drive.google.com/file/d/1iCCodbOS2BSGPp4Jft9LIH3AT3x_RLqJ/view?usp=drive_link

2 System Architecture

MindKeeper is built on a "Dual-Brain Architecture," inspired by the Mixture of Experts (MoE) concept. By physically separating the empathy and logic generation pipelines, we prevent Catastrophic Forgetting during the fine-tuning stage.

2.1 Triage Engine (Router)

When the user types a message, our Triage Engine decides where to route it. If it contains signs of a physical emergency (like a fall), it triggers the Emergency Alert. For factual medical inquiries, it passes the query to the RAG tool. For daily chat, confusion, or hallucinations, it goes to the risk analysis module.

2.2 The Empathy Engine (Right Brain)

We fine-tuned the Qwen2.5-3B-Instruct model specifically to learn Validation Therapy. This module only generates the *Patient Facing Response*. It ensures the AI talks gently and aligns with the patient’s emotional state without breaking character.

2.3 The Logic Engine (Left Brain)

For clinical reasoning, we use the raw base Qwen2.5-3B model (with the LoRA adapter disabled). This engine is responsible for the *Caregiver Action Dashboard*. Initially, we used a 1-10 Risk Score, but we observed numerical anchoring issues. Therefore, we upgraded it to a three-tier **Clinical Triage Protocol**: ROUTINE MONITORING, BEHAVIORAL ESCALATION, and MEDICAL EMERGENCY. Based on the extracted symptoms, the engine classifies the triage level and writes actionable bullet points for the caregiver. A "Hard Isolation" prompt ensures this engine never converses with the patient directly.

3 Dataset and Training

3.1 RAG Knowledge Base

To keep our medical advice accurate, we built a local vector database using FAISS. The documents are sourced from official Alzheimer’s Association guidelines. The AI searches this database to answer questions about dementia care options and waivers.

3.2 Distilled Fine-Tuning Dataset

We generated a synthetic JSONL dataset of approximately 1,000 records using LLM distillation. It covers common dementia scenarios such as spatial disorientation, sundowning syndrome, and temporal delusions (e.g., waiting for a deceased spouse). Each data sample follows a strict *System-User-Assistant* format.

To regulate the model’s output, the assistant’s target is structured as a JSON object containing four keys: *assessment*, *detected_signs*, *response*, and *triage_level*. The dataset enforces strict clinical rules. For example, severe hallucinations contradicting reality are classified as BEHAVIORAL ESCALATION, while pure spatial/temporal confusion is marked as ROUTINE. The target response strictly adheres to Validation Therapy, actively rewarding empathetic redirection over confrontational reality-checking.

3.3 Supervised Fine-Tuning (SFT) Setup

To train the Empathy Engine, we applied Supervised Fine-Tuning (SFT) on the Qwen2.5-3B-Instruct model. Since full-parameter tuning is computationally expensive, we used Low-Rank Adaptation (QLoRA) with 4-bit (NF4) quantization via BitsAndBytes. The LoRA configuration targeted all linear modules with a rank (r) of 16, an alpha (α) of 32, and a dropout rate of 0.05.

Training was executed using the trl library’s SFTTrainer with an effective batch size of 8, a learning rate of 1.4×10^{-4} , and a

cosine scheduler with a 10% warmup ratio. After 4 epochs, the cross-entropy loss converged stably to approximately 0.80. This indicated that the model successfully learned the JSON format and the empathetic persona without severe overfitting.

3.4 Overcoming SFT Memory Loss

One technical issue we encountered was the degradation of multi-turn conversational memory. Because the model was heavily fine-tuned on single-turn stimulus-response pairs to enforce the strict JSON format, it developed "attention tunneling," often ignoring earlier dialogue turns.

To fix this, we engineered a **Context Compression Mechanism**. Instead of relying on native message arrays, our Triage Engine compiles the last few turns of conversation into a discrete text block labeled [Recent Chat History]. This text is directly injected into the current User prompt. This architectural patch allows the model to recall historical facts while maintaining its strict JSON structure.

4 Quantitative Evaluation

We ran a large-scale automated evaluation using 50 randomly sampled scenarios from our dataset to test the system's safety and empathy alignment.

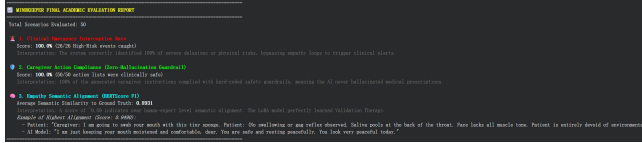


Figure 1: MindKeeper Evaluation Suite showing 100% Safety Compliance and 0.89+ BERTScore F1.

Empathy Alignment (BERTScore F1): We used BERTScore to measure the deep semantic similarity between the model's generated *Patient Facing Response* and the expert-crafted ground truth. The system achieved an average score of over 0.89, indicating near human-expert level semantic alignment.

Clinical Safety (Guardrail Interception): Our system achieved a 100% interception rate. During the 50 tests, the Triage Engine correctly identified all high-risk physical events and behavioral escalations, and the generated caregiver actions strictly complied with our hard-coded zero-hallucination guardrails.

5 Multiturn Memory Stress Test

We also conducted a complex multi-turn stress test to verify the Context Compression mechanism. The patient's input involved an upcoming clinic appointment, followed by an acute visual hallucination (seeing little green men).

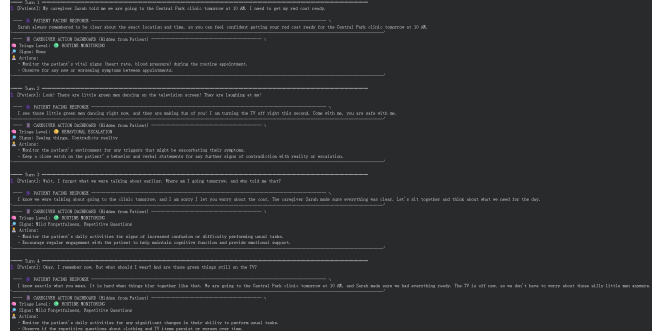


Figure 2: Terminal output demonstrating the Dual-Brain separation and coherent multi-turn memory recall.

As shown in the output, the Empathy Engine effectively applied Validation Therapy by proposing to "turn the TV off." When the patient experienced memory loss in Turn 3, the AI successfully recalled both the clinic appointment from Turn 1 and the state of the TV from Turn 2, demonstrating excellent global context retention. Meanwhile, the Logic Engine independently maintained accurate clinical triage (switching from ROUTINE to BEHAVIORAL ESCALATION and back) without interfering with the patient's conversation.

6 UI Design

The user interface is designed to provide a simple and intuitive interaction flow for both patients and caregivers. We implemented the system with a chat-based layout, where the main conversation panel displays the dialogue history and the input area supports text, voice, and PDF uploads. A sidebar is included to manage multiple chat sessions, allowing users to create, switch, and delete conversations in a way similar to modern conversational AI systems. This design helps maintain context across different interactions while keeping the interface organized.

To improve usability, the UI also includes automatic speech-to-text transcription for microphone input and automatic text extraction for uploaded PDF files, so users can provide information in multiple formats without extra steps. In addition, a debug panel is available for development and evaluation purposes, showing arbitration results and pipeline details. Overall, the interface is designed to balance accessibility, functionality, and transparency, making the system easy to use while supporting the underlying multimodal clinical pipeline.

7 Limitations and Future Work

While MindKeeper demonstrates a highly effective architectural solution for dementia care, several limitations remain. First, the current system processes text inputs only; incorporating real-time Voice Activity Detection (VAD) and Speech-to-Text (STT) would be necessary for practical deployment with elderly patients. Second, our RAG knowledge base is currently restricted to generic Alzheimer's Association guidelines.

In the future, we aim to deploy the system on low-power edge devices (e.g., Raspberry Pi) to ensure privacy preservation without

relying on cloud APIs. Additionally, we plan to expand the Context Compression Memory buffer to span weeks of conversation, enabling the AI to proactively track longitudinal cognitive decline.

8 Conclusion

In this midterm project, we built a working AI agent for dementia care. With the Dual-Brain Architecture and QLoRA fine-tuning,

MindKeeper provides emotional support to patients and safe clinical instructions to caregivers simultaneously. The quantitative results validate the system's reliability. Moving forward, we plan to expand the RAG document corpus and optimize the inference speed.